TCG-FlexiMetal 35 Foil (v9.1)

Deep Materials

Product Description

TCG-FlexiMetal 35 (v9.1) Foil is an advanced metallic thermal interface material engineered to deliver ultra-low thermal resistance without the limitations of dry metal foils or liquid metal TIMs. Its viscosity-controlled metallic structure enables wetting and conformal contact across warped and non-coplanar surfaces, preserving interface integrity over time. The material minimizes mechanical stress on devices by achieving effective thermal contact at relatively low compression force, making it suitable for fragile dies and large-area packages. Supplied as a foil structure, FlexiMetal supports multiple integration approaches, including pre-attachment and bonded interfaces, for scalable manufacturing and consistent system-level performance. FlexiMetal contains no organic material and requires no burn-in as compared to PCM.

Key Features

- 100% non-organic thermal interface delivering ultra-low thermal resistance, even at increased bond line thicknesses to accommodate die warpage and flatness variations.
- Engineered high-viscosity metallic formulation minimizes flow and eliminates the leakage and electrical shorting risks associated with liquid metal TIMs.
- Enhances both steady-state and transient thermal performance by rapidly transferring heat from silicon to the thermal solution.
- Designed for simplified handling and compatibility with standard assembly processes.
- Supports bonding to heat sinks or lid surfaces.
- Alternative paste version supports screen printing.
- Enables simplified field rework and replacement compared to liquid metal interfaces, without requiring specialized handling procedures or containment controls.

Applications

- Interfaces with significant die warpage or surface non-uniformity where conventional TIMs cannot maintain low thermal resistance while minimizing resistance.
- Cooling of high-power AI chips with MCM
- Cooling of server CPUs
- **Replacements for liquid metal TIMs where leakage, migration, or electrical risks are unacceptable**
- Thermal interfaces requiring consistent performance at increased bond line thickness
- Designs requiring flexible integration, including foil, pre-attachment, or bonded configurations

Application Guidelines

- Avoid exposure to open flame and liquid water.
- Ensure application surfaces are clean.
- Can be pre-attached to the heat sink and shipped as an integrated assembly to the end customer.
- Application support is available to optimize handling, placement, pre-attachment, and pressure conditions.

Storage and Shelf Life

- Store at 8℃ to 35℃ in a dry environment.
- Shelf life is 12 months from the date of manufacture.
- Please contact us for application notes and further instructions as this is a new material.

Property & Units	Value
Thermal conductivity (W/(m·K))	35
Total foil assembly thickness (µm)	130
Density (g/cm^3)	Thickness Dependent
Appearance	Silvery
Active heat transfer area (FlexiMetal) (mm^2)	Per Customer Requirement
Polyamide frame width (mm)	3 - 5 (Customizable)
Young's modulus (25 °C – material in solid state) (MPa)	0.25
Melting point (°C)	~ 10
Boiling point (°C)	> 2200
Thermal impedance at 130 µm BLT (°C·cm²/W)	~ 0.025

Note: Data shown are preliminary and may change. FlexiMetal-9.1 is an advanced thermal interface technology under active development and is covered by issued and pending patents in the U.S. and worldwide.

The data provided are typical values derived from laboratory testing and are intended for reference only. They should not be used as specifications or as a basis for design without independent validation.